

HANDWRITTEN DIGIT CLASSIFICATION USING DEEP LEARNING

Internship Project Report

1. Project Title:

Handwritten Digit Classification Using Deep Learning

2. Objective:

The objective of this project is to develop a deep learning model capable of classifying handwritten digit images from 0 to 9. The system aims to automatically recognize digit patterns accurately using neural network techniques.

3. Problem Statement:

Handwritten digit recognition is a widely used application of deep learning and image classification. Due to different handwriting styles, manual recognition can be difficult. This project solves the problem by building a deep learning model that predicts the correct digit from image data.

4. Dataset Description:

- The project uses the Digits Dataset from Scikit-learn.

Dataset Features:

- Total Samples: 1797
- Image Size: 8×8 pixels
- Classes: Digits from 0 to 9

Input Features:

- Pixel intensity values of handwritten digit images

Target Variable:

Digit label (0–9)

5. Methodology

5.1 Data Preprocessing:

- Dataset was loaded using Scikit-learn
- Image data was converted into numerical feature vectors
- Dataset was split into training and testing sets

5.2 Model Selection:

The following deep learning model was used:

- MLPClassifier (Multi-Layer Perceptron)

5.3 Training Process:

- The deep learning model was trained using training data
- Predictions were generated on test data
- Model accuracy was evaluated

6. Evaluation Metrics

The model performance was evaluated using:

- Accuracy Score
- Prediction Results
- Classification Performance

7. Results

- Deep Learning Model Accuracy: Approximately 95%–98%
- Best Performing Model: Multi-Layer Perceptron (MLP Classifier)

The model successfully classified handwritten digit images with high accuracy.

8. Visualization

The following visualizations were used:

- Sample handwritten digit images
- Predicted digit outputs
- Accuracy results

These visualizations helped in understanding the model's prediction performance and dataset patterns.

9. Future Improvements

- Use larger real-world image datasets
- Implement Convolutional Neural Networks (CNN)
- Improve model architecture
- Deploy project using Flask or Streamlit
- Improve accuracy using hyperparameter tuning

10. Tools & Technologies Used

1. Python
2. Scikit-learn
3. NumPy
4. Matplotlib
5. Jupyter Notebook

Conclusion

The deep learning model successfully classified handwritten digit images with high accuracy. The project demonstrates the practical application of deep learning techniques in image classification and pattern recognition tasks.

This system shows how neural networks can be used effectively for handwritten digit recognition and intelligent prediction systems.

Author

Internship Project – Deep Learning Task 2